MIL-T-28764(EC)
AMENDMENT 5
30 May 1974
SUPERSEDING 1/
~~AMENDMENT 4~~
20 October 1972

MILITARY SPECIFICATION
TEST SET, RADAR, AN/UPM-136( )

This amendment forms a part of Military Specification MIL-T-28764(EC) dated 30 April 1971.

Page 1

2.1, SPECIFICATIONS, MILITARY: Delete "MIL-C-17/84 Cable, Radio Frequency, Coaxial, RG-223/U" and substitute "MIL-C-17/75 Cable, Radio Frequency, Coaxial, RG-214/U".

2.1, SPECIFICATIONS, MILITARY: Add "MIL-C-28777/3 Cable, Assembly, Electronic Test Equipment (3 wire, 3 prong, 125 volt, Molded) Grounding Plug Connector, and 90 Degree Molded MIL-C-5015 Connector, 60 Hertz" and "MIL-C-45662 Calibration System Requirements".

Page 2

2.1, STANDARDS, MILITARY, after "MIL-STD-810", add:

"MIL-STD-1345	Data, Measurement, In Support of Maintenance, Calibration and Repair of Electronic Equipment"

2.1, PUBLICATIONS: Delete "NAVSHIPS 0969-125-0250 Navy Electronic Systems Command Preferred GPETE List" and substitute "MIL-STD-1364. Preferred General Purpose Electronic Test Equipment".

Page 3

3.3, line 1: Delete "Technical Standards for the Air Traffic Control Radar Beacon System (ATCRBS)/IFF/Mark XII Systems (AIMS)" and substitute "US National Standards for Command System Component Characteristics for the IFF Mark X (SIF) Air Traffic Control Radar Beacon Systems SIF/ATCRBS".

1/ CHANGES FROM PREVIOUS ISSUE. THE OUTSIDE MARGINS OF THIS DOCUMENT HAVE BEEN MARKED "#" TO INDICATE WHERE CHANGES (DELETIONS, ADDITIONS, AND SO FORTH) FROM THE PREVIOUS ISSUE HAVE BEEN MADE. THIS HAS BEEN DONE AS A CONVENIENCE ONLY AND THE GOVERNMENT ASSUMES NO LIABILITY WHATSOEVER FOR ANY INACCURACIES IN THESE NOTATIONS. BIDDERS AND CONTRACTORS ARE CAUTIONED TO EVALUATE THE REQUIREMENTS OF THIS DOCUMENT BASED ON THE ENTIRE CONTENT AS WRITTEN IRRESPECTIVE OF THE MARGINAL NOTATIONS AND RELATIONSHIP TO THE LAST PREVIOUS ISSUE.

FSC 6625

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Page 5

3.6.1.5, line 1: Delete "3.6.1.1" and substitute "3.6.1.2"; and line 2: Delete "3.7.8" and substitute "3.8.8".

3.6.1.10, last sentence: Add "under normal operating conditions".

Page 6

3.6.1.19: Delete and substitute the following:

"3.6.1.19 Instrumentation requirements. Instrumentation requirements applicable to this test set shall be as follows:

"3.6.1.19.1 Circuit test points. Circuit test point accesses shall be provided which comply with the criteria outlined in MIL-STD-415. The following types of tests shall be considered for determining the test points:

(a) Signal quality
(b) Circuit performance
(c) Calibration
(d) System alignment
(e) Installation and check-out
(f) Planned maintenance systems tests
(g) Line replaceable unit (LRU) fault location
(h) Shop replaceable part (SRP) fault isolation

Test points for calibration shall be provided external to the device under test (DUT), except where it would affect the calibration. In addition, test points shall be provided to the LRU interconnection connector for fault, isolation of SRPs (non-repairables) or pertinent functional circuits within the LRU, whichever is the most desirable to fault isolate (test) by shop programmable test instrumentation."

Add the following new paragraphs:

"3.6.1.20 Support test instrumentation. Support test instrumentation shall be as follows:

"3.6.1.20.1 General purpose electronic test equipment (GPETE). GPETE shall be selected from MIL-STD-1364. Approval of the utilization of GPETE not on this list for support testing shall be requested in writing, with justification consisting of parameters to be measured, techniques, accuracies, periodicity of tests, where tests are to be made, and so forth.".

"3.6.1.20.2 Line operation and maintenance test support. The selection of GPETE shall be determined to perform the following tests:

(a) Signal quality

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(b) Circuit performance
(c) Alignment
(d) Tuning
(e) Peaking
(f) Adjustments
(g) Calibration
(h) LRU fault location

3.6.2.1: Delete and substitute:

"3.6.2.1 Size The size and mounting dimensions of the equipment, including top mounted accessory case, shall not exceed 15 inches (37.5 cm) high by 14 inches (35 cm) wide and by 24 inches (60 cm) deep."

3.6.2.4: Delete "Panel Protection." and substitute "Panel protection and layout." and add "The front panel layout shall use figure 2 as a guide."

Page 7

3.7.1, line 2: Delete "16400" and substitute "MIL-E-16400".

3.8, (a): Delete "3.7.1" and substitute "3.8.1".
(b): Delete "3.7.2" and substitute "3.8.2".
(c): Delete "3.7.5" and substitute "3.8.5".

3.8.2.1, last line: Delete "3.7.2.5.1." and substitute "3.8.2.5.1."

3.8.2.2, last sentence: Delete "3.7.2.2.1" and substitute "3.8.2.2.1" and delete "3.7.2.2.6." and substitute "3.8.2.2.6."

Page 8

3.8.2.3, line 2: Delete "(see 3.7.3.5.1)" and substitute "(see 3.8.3.5.1)".

Page 10

3.8.3.7.1, line 2: Delete "test (3.8.3.2.1)" and substitute "test (3.8.3.2.3)".

Page 11

3.8.4.2: Delete and substitute:

"3.8.4.2 Marker switch. A three position (1.00 μsec/OFF/1.45 μsec) crystal marker switch shall be located on the front panel of the unit. Markers shall be provided with a spacing of 1.0 ± 0.02 μsec or 1.45 ± 0.02 μsec. The markers shall be continuous and synchronous with the zero delay trigger specified in 3.8.2.5.1. The width of the 1.0 μsec markers shall not exceed 0.05 μsec and the width of the 1.45 μsec markers shall not exceed 0.20 μsec."

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3.8.5.1: Delete and substitute:

"3.8.5.1 Accessory case. A slide door type, front opening, accessory case capable of being mounted on top of the test set and having the capacity to hold the accessories specified herein, plus technical manuals, if required, shall be provided."

3.8.5.2, line 2: Delete "Table I" and substitute "Table V". Also, Table I, Video test probe characteristics: Delete "Table I" and substitute "Table V".

3.8.5.3: Delete and substitute:

"3.8.5.3 High power rf cable. One ten foot length of rf cable type RG-214/U in accordance with MIL-C-17/75, having a M39012/6-0015 cabled-plug in accordance with MIL-C-39012/6 on one end and a M39012/01-0005 in accordance with MIL-C-39012/1 on the other end shall be provided. This cable shall be marked to indicate its attenuation."

3.8.5.6, line 6: Delete "RG-223/U cable in accordance with MIL-C-17/84" and substitute "RG-214/U cable in accordance with MIL-C-17/75."

3.8.8: Delete and substitute:

"3.8.8 Primary power cable. Cable assembly shall be furnished in accordance with MIL-C-28777/3."

4.3, last line: Delete "table I." and substitute "table II.".

Table I, heading: Delete "Table I." and substitute "Table II.".

In "Table I. First article inspection (continued)", delete "Table I." and substitute "Table II." Under "Examination or test" column, add: "Support maintenance test procedure verification". Under "Requirement paragraph" and "Test method paragraph", add "3.6.1.19" and "4.7.8", respectively.

4.4.1, last line: Delete "table II." and substitute "table III.".

In Table II heading, delete "II" and substitute "III".

4.4.2, line 2, delete "table III" and substitute "table IV".

In table III heading, delete "III" and substitute "IV".

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Page 17

4.4.3, line 1, delete "table IV" and substitute "table V".

Table IV heading, delete "IV." and substitute "V.".

Page 19

After 4.7.7, add the following new paragraphs:

"4.7.8 Support maintenance test procedure (SMTP) verification. The procedures and techniques developed for performance of all types of maintenance tests for each level (organizational, intermediate, depot, special repair facilities, and calibration facilities) shall be verified prior to submittal of the final data required by MIL-STD-1345, Navy training instructions, and final Navy technical manuals (see 6.3).

"4.7.8.1. The support maintenance test procedure verification shall be performed by a qualified technician selected by the Government. The technician may be selected from either Government or contractor personnel.

"4.7.8.2. The test procedures used in the SMTP verification may be preliminary procedures, pamphlets or manuals (not necessarily in the technical manual format), or preliminary manuals (in the required format).

"4.7.8.3. Test equipment used to verify the SMTP shall be selected from MIL-STD-1364. Test equipment not specified in MIL-STD-1364 may be used only upon approval of the command or agency concerned, in response to the contractor's written request for such approval (see 6.3).

"4.7.8.4. The SMTP verification shall be performed on one of the first article models and an item selected at random from the production line.

"4.7.9 Support maintenance test procedure verification. Test equipment support maintenance test procedure verification shall be as follows:

(a) The technician will be given a complete set of boards, modules, and parts, all of which are completely unaligned.

(b) The technician shall align each of the boards or modules and verify that each board or module is operating properly. These tests shall be performed with the boards or module on the bench (module removed from the equipment).

(c) The technician shall install the boards or modules in the test equipment and align the test equipment as specified in the manual.

(d) The equipment shall then be calibrated by a calibration laboratory as specified in the instrument calibration procedure (ICP). The calibration facility selected shall conform to the requirements of MIL-C-45662."

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6.3, add:

"(1) Request for approval of non-preferred GPETE (4.7.8.3).
(m) Electromagnetic interference (see 3.6.1.16)."

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